

Linux Challenges

Task 1: Launch one aws ec2 and install jenkins.

Objective:

Launch an AWS EC2 instance and install Jenkins.

Purpose:

To set up Jenkins on an AWS EC2 server for CI/CD automation.

Implementation:

1. Launched an Ubuntu EC2 instance and connected using SSH.
2. Installed Java 21.
3. Installed Jenkins.
4. Started Jenkins using systemctl.
5. Verified the Jenkins service status successfully.

```
ubuntu@ip-172-31-24-128: ~
$ cd ~/Downloads
$ chmod 400 jenkins-key.pem
$ ssh -i jenkins-key.pem ubuntu@34.233.126.119
ssh: connect to host 34.233.126.119 port 22: Connection timed out
$ AC
$ ssh -i jenkins-key.pem ubuntu@34.233.126.119
ssh: connect to host 34.233.126.119 port 22: Connection timed out
$ ssh -i jenkins-key.pem ubuntu@34.233.126.119
The authenticity of host '34.233.126.119 (34.233.126.119)' can't be established.
ED25519 key fingerprint is SHA256:KqnsVPM4ckvkZr/gxvBnkHOFqbiqz14tzXfBzky7g.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.233.126.119' (ED25519) to the list of known hosts
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Aug 10 10:30:49 UTC 2026

System load: 0.04          Temperature: -273.1 C
Usage of /: 30.5% of 6.61GB Processes: 116
Memory usage: 25%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.24.128

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-128:~$ java -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-24-128:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-08-10 10:38:13-- https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:95::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring.asc 100%[=====]

2026-08-10 10:38:13 (25.0 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]

ubuntu@ip-172-31-24-128:~$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] https://pkg.jenkins
/etc/apt/sources.list.d/jenkins.list > /dev/null
ubuntu@ip-172-31-24-128:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:7 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Get:8 https://pkg.jenkins.io/debian-stable binary/ Packages [31.4 kB]
```

```
ubuntu@ip-172-31-24-128:~$ sudo apt install fontconfig openjdk-21-jre -y
Fontconfig is already the newest version (2.17.1-1ubuntu1).
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  openjdk-21-jre

Installing dependencies:
  ibus-gtk3 libcolord2 libepoxy0 libgtk-3-0t64 libgtk-3-bin libgtk-3-common libwayland-cursor0 libwayland-egl1 openjdk-21-jre-headless

Suggested packages:
  colord gvfs libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 5
  Download size: 52.0 MB
  Space needed: 221 MB / 3713 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 ibus-gtk3 amd64 1.5.34-rc2-1 [19.9 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libcolord2 amd64 1.4.8-3 [151 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libepoxy0 amd64 1.5.10-2build1 [256 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libwayland-cursor0 amd64 1.24.0-2 [10.8 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libwayland-egl1 amd64 1.24.0-2 [6264 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libgtk-3-common all 3.24.52-0ubuntu1 [1424 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libgtk-3-0t64 amd64 3.24.52-0ubuntu1 [3085 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libgtk-3-bin amd64 3.24.52-0ubuntu1 [74.3 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 openjdk-21-jre-headless amd64 21.0.11-10-1-26.04.2 [46.8 MB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 openjdk-21-jre amd64 21.0.11-10-1-26.04.2 [227 kB]
Fetched 52.0 MB in 1s (93.9 MB/s)
Selecting previously unselected package ibus-gtk3:amd64.
(Reading database ... 132337 files and directories currently installed.)
Preparing to unpack .../0-ibus-gtk3_1.5.34-rc2-1_amd64v3.deb ...
Unpacking ibus-gtk3:amd64 (1.5.34-rc2-1) ...
Selecting previously unselected package libcolord2:amd64.
Preparing to unpack .../1-libcolord2_1.4.8-3_amd64v3.deb ...
Unpacking libcolord2:amd64 (1.4.8-3) ...
Selecting previously unselected package libepoxy0:amd64.
Preparing to unpack .../2-libepoxy0_1.5.10-2build1_amd64v3.deb ...
Unpacking libepoxy0:amd64 (1.5.10-2build1) ...
Selecting previously unselected package libwayland-cursor0:amd64.
Preparing to unpack .../3-libwayland-cursor0_1.24.0-2_amd64v3.deb ...
Unpacking libwayland-cursor0:amd64 (1.24.0-2) ...
Selecting previously unselected package libwayland-egl1:amd64.
Preparing to unpack .../4-libwayland-egl1_1.24.0-2_amd64v3.deb ...
Unpacking libwayland-egl1:amd64 (1.24.0-2) ...
```

```
ubuntu@ip-172-31-24-128:~$ sudo apt install jenkins -y
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  jenkins

Installing dependencies:
  net-tools

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 5
  Download size: 101 MB
  Space needed: 102 MB / 3815 MB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 net-tools amd64 2.10-2ubuntu1 [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.568.2 [101 MB]
Fetched 101 MB in 1s (73.1 MB/s)
```

```
ubuntu@ip-172-31-24-128:~$
No containers need to be restarted.

User sessions running outdated binaries:
  ubuntu @ session #1: sshd-session[1156,1273]
  ubuntu @ user manager: (sd-pam)[1163]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-24-128:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-24-128:~$ readlink -f $(which java)
/usr/lib/jvm/java-21-openjdk-amd64/bin/java
ubuntu@ip-172-31-24-128:~$ sudo systemctl start jenkins
ubuntu@ip-172-31-24-128:~$ sudo systemctl status jenkins
● Jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-10 10:48:44 UTC; 14s ago
     Invocation: a103944417cd4e709895c40b2830bf90
       Main PID: 25658 (java)
         Tasks: 41 (limit: 227)
        Memory: 314.9M (peak: 319.3M)
           CPU: 20.894s
      CGroup: /system.slice/jenkins.service
              └─25658 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]
Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]
Aug 10 10:48:39 ip-172-31-24-128 jenkins[25658]: [LF]
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.126+0000 [id=34] INFO Jenkins.InitReactorRunner$1.onAttained: Completed initialization
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.171+0000 [id=24] INFO Hudson.lifecycle.Lifecycle$onReady: Jenkins is fully up and running
Aug 10 10:48:44 ip-172-31-24-128 system[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.360+0000 [id=50] INFO hudson.DownloadService$Downloadable#load: Obtained the updated data
Aug 10 10:48:44 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:44.361+0000 [id=50] INFO hudson.util.Retrier$start: Performed the action check updates sen
Aug 10 10:48:49 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:48:49.250+0000 [id=69] WARNING hudson.DiskSpaceMonitorDescriptor#markNodeOfflineorOnline: Making 2
lines 1-21/21 (END)

[1] Stopped
ubuntu@ip-172-31-24-128:~$ sudo ss -ltnp | grep 8080
LISTEN 0 50 *:* users:((("java",pid=25658,fd=9))
ubuntu@ip-172-31-24-128:~$ cat /var/lib/jenkins/secrets/initialAdminPassword
cat: /var/lib/jenkins/secrets/initialAdminPassword: Permission denied
ubuntu@ip-172-31-24-128:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
9a24f423e564d348bb5cca6e3f83eb3
ubuntu@ip-172-31-24-128:~$ sudo systemctl stop jenkins
ubuntu@ip-172-31-24-128:~$ sudo systemctl status jenkins
● Jenkins.service - Jenkins Continuous Integration Server
```

devops-cl | benefits-cl | Techie Ho... | Instance cl... | Inbox - va... | Linux Com... | Setup Wi... | Ask Gemini

Not secure 34.233.126.119:8080

Getting Started

Create First Admin User

Username

vmshd

Password

Confirm password

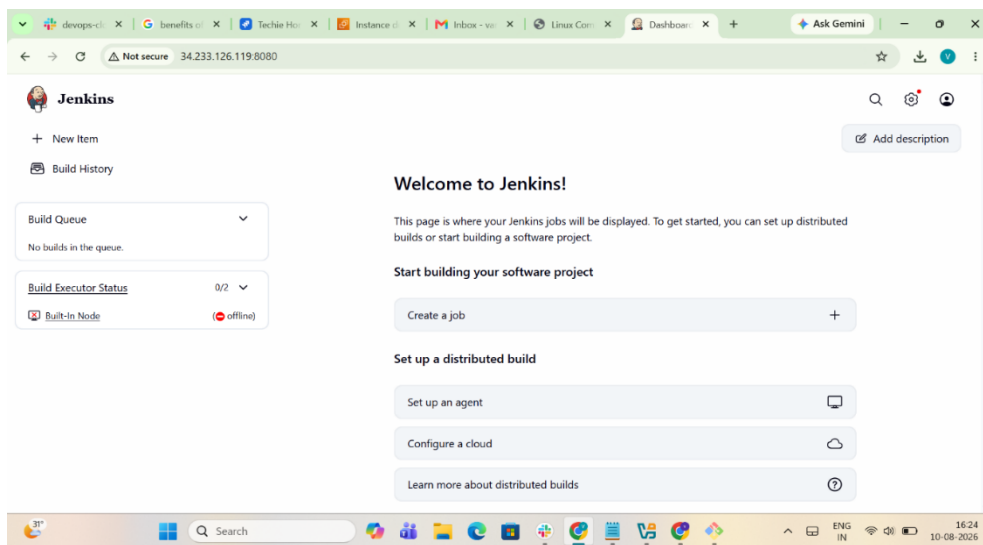
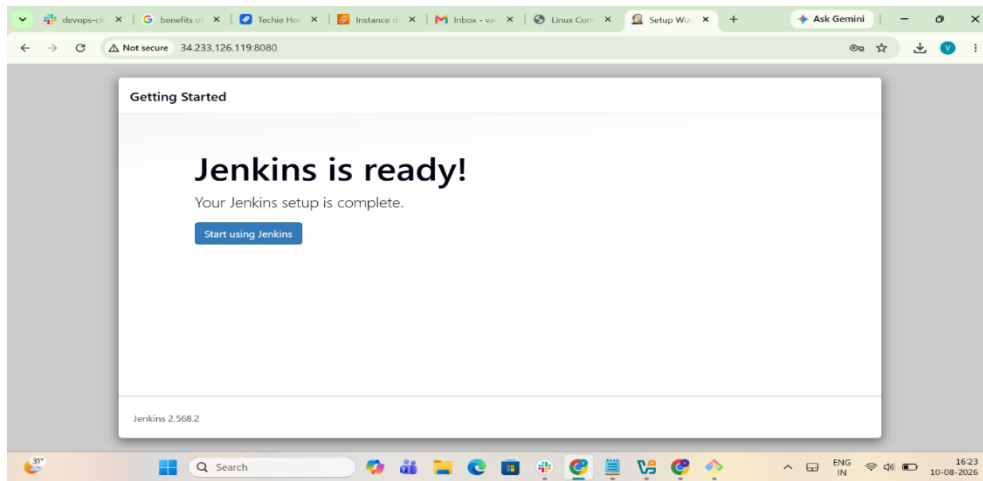
Full name

vmshikumar

Jenkins 2.568.2

Skip and continue as admin

Save and Continue



```
ubuntu@ip-172-31-24-128:~$ sudo systemctl stop jenkins
ubuntu@ip-172-31-24-128:~$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-08-10 10:55:46 UTC; 6s ago
     Duration: 7min 1.772s
  Invocation: a103944417cd4e709895c40b2830bf90
    Process: 25658 ExecStart=/usr/bin/jenkins (code=exited, status=143)
   Main PID: 25658 (code=exited, status=143)
    Status: "Jenkins stopped"
   Mem peak: 523.3M
      CPU: 1min 2.194s

Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.030+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpDisconnectComputers: Starting n
Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.036+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpShutdownPluginManager: Stopping
Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.040+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpPersistQueue: Persisting build
Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.051+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpAwaitDisconnects: waiting for n
Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.051+0000 [id=26] INFO hudson.lifecycle.Lifecycle#cleanStatusUpdate: Jenkins stopped
Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.060+0000 [id=26] INFO winstone.Logger#logInternal: Control thread shutdown successf
Aug 10 10:55:46 ip-172-31-24-128 jenkins[25658]: 2026-08-10 10:55:46.060+0000 [id=26] INFO winstone.Logger#logInternal: Jetty shutdown successfully
Aug 10 10:55:46 ip-172-31-24-128 systemd[1]: jenkins.service: Deactivated successfully.
Aug 10 10:55:46 ip-172-31-24-128 systemd[1]: Stopped jenkins.service - Jenkins Continuous Integration Server.
Aug 10 10:55:46 ip-172-31-24-128 systemd[1]: jenkins.service: Consumed 1min 2.194s CPU time over 7min 13.971s wall clock time, 523.3M memory peak.
```

Conclusion:

Successfully launched an AWS EC2 instance and installed and configured Jenkins.

Task 2: Setup Apache Tomcat on Ubuntu Virtual Machine.

Objective:

Set up and run Apache Tomcat on an Ubuntu Virtual Machine.

Purpose:

To deploy and run Java-based web applications using Apache Tomcat.

Implementation:

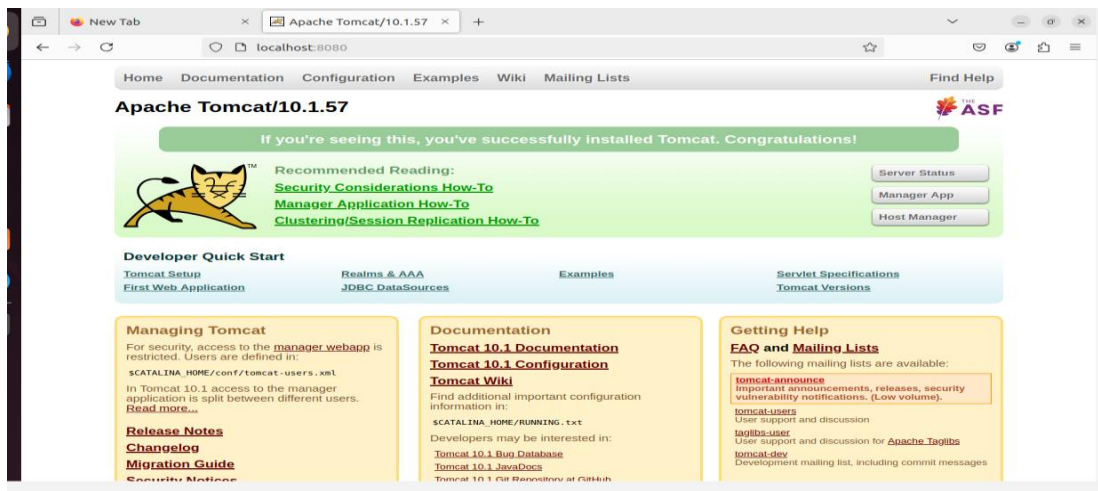
1. Checked Tomcat service status: `sudo systemctl status tomcat`
2. Verified Tomcat is **active (running)**.

3. Verified Tomcat is listening on port 8080.

4. Verified the server startup successfully through the service logs.

```
root@vanshikumar-VirtualBox:~# cd /opt
vanshikumar@vanshikumar-VirtualBox:/opt$ sudo wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
--2026-08-10 15:21:14-- https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.46/bin/apache-tomcat-10.1.46.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-10 15:21:15 ERROR 404: Not Found.
```

```
apache-tomcat-10.1.57.tar.gz tomcat
vanshikumar@vanshikumar-VirtualBox:/opt$ ls /opt/tomcat
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
vanshikumar@vanshikumar-VirtualBox:/opt$ ll
total 14076
drwxr-xr-x 3 root root 4096 Aug 10 15:26 ./
drwxr-xr-x 20 root root 4096 Aug 4 06:22 ../
-rw-r--r-- 1 root root 14400273 Jul 3 18:43 apache-tomcat-10.1.57.tar.gz
drwxr-xr-x 9 root root 4096 Aug 10 15:25 tomcat/
vanshikumar@vanshikumar-VirtualBox:/opt$ sudo chmod +x /opt/tomcat/bin/*.ssh
chmod: cannot access '/opt/tomcat/bin/*.ssh': No such file or directory
vanshikumar@vanshikumar-VirtualBox:/opt$ sudo chmod +x /opt/tomcat/bin/*.sh
chmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
vanshikumar@vanshikumar-VirtualBox:/opt$
```



Conclusion:

Successfully configured and verified Apache Tomcat running on Ubuntu VM on port 8080.

Task 3: Setup Docker Desktop on your local windows.

Objective:

Install and configure Docker Desktop on a local Windows machine.

Purpose:

To set up a local containerization environment for developing and running Docker applications.

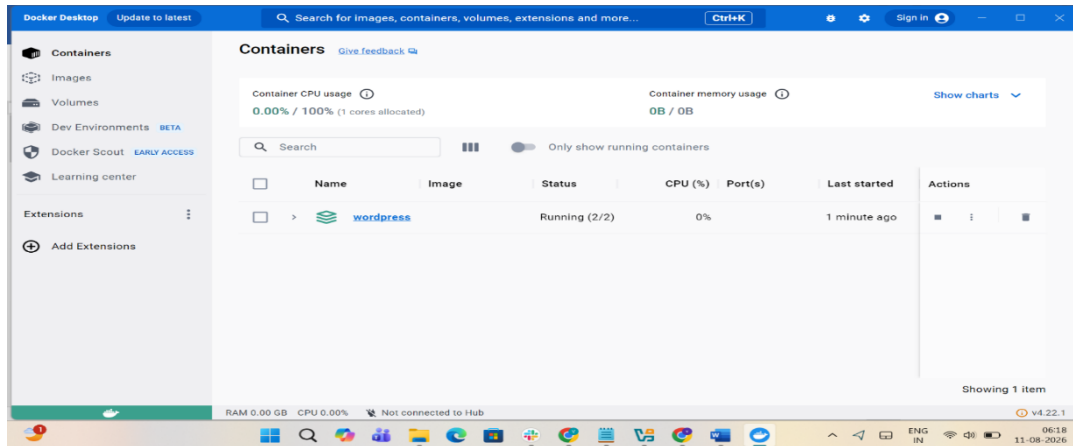
Implementation:

1. Downloaded and installed Docker Desktop.
2. Enabled the required Windows virtualization features.
3. Started Docker Desktop successfully.
4. Verified the installation: `docker --version`

```
MINGW64/c/Users/91738

91738@IDEAPAD MINGW64 ~
$ docker --version
Docker version 24.0.5, build ced0996

91738@IDEAPAD MINGW64 ~
$ |
```



Conclusion:

Successfully installed and configured Docker Desktop on Windows.